

HASHWANTH GHANTA

M.Sc. Digital Engineering Candidate | Software Developer | Unity 3D · iOS · Backend

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PROFESSIONAL SUMMARY

M.Sc. Digital Engineering student at Otto-von-Guericke-Universität Magdeburg with a B.Tech in Electronics and Communication Engineering. Experienced in C# software development, Unity 3D simulation engineering, SwiftUI iOS app development, Python data science, and web development. Combines a strong foundation in embedded systems and signal processing with modern software engineering practice in C#, Python, Swift, and JavaScript. Available May 2026 for a Werkstudent or internship position in Germany.

TECHNICAL SKILLS

Programming Languages: C, C#, Python, Swift, JavaScript, HTML5, CSS3

Simulation and Game Development: Unity 3D, MonoBehaviour, ScriptableObjects, Event Systems, Particle Systems

iOS and Mobile Development: SwiftUI, UIKit, Swift Charts, XCTest, MVVM, Xcode, iOS

Web Development: HTML5, CSS3, JavaScript (ES6+), Responsive Design, REST APIs, EmailJS, GitHub Pages

Data and Visualisation: Python, Data Dashboards, Real-time Data Rendering, NumPy, Pandas

3D and Design: Blender (3D Modelling), Adobe Premiere Pro, Canva, UI/UX Design

Tools and DevOps: Git, GitHub, GitHub Actions, CI/CD, VS Code, Visual Studio, Xcode

Domains: Embedded Systems, Signal Processing, Simulation Engineering, IoT, Data-Driven Development, Accessibility (WCAG 2.2 AA)

EDUCATION

Otto-von-Guericke-Universität Magdeburg (OVGU), Magdeburg, Germany

M.Sc. Digital Engineering April 2025 – Present

- Focus: data-driven software development, simulation engineering, embedded systems, human-computer interaction
- Coursework: Advanced Software Engineering, Digital Systems Design, Human-Computer Interaction

VIT-AP University, Amaravati, India

B.Tech, Electronics and Communication Engineering August 2020 – June 2024

- GPA: 7.72 / 10.0
- Coursework: Signal Processing, Embedded Systems, Programming in C and C++, Communication Theory

PROJECTS

CO₂ Carbon Capture Process Simulation — Unity 3D, C# 2025 – Present

- Developed a fully interactive Unity 3D educational simulation of a seven-phase post-combustion CO₂ carbon capture process.

- Designed a custom C# architecture with real-time dynamic UI panels for step-by-step guided learning.
- Implemented particle-based fluid flow animations and a responsive 3D environment to visualise industrial CCS technology.
- Built modular scene management and an event-driven UI system supporting reusable learning modules.
- Live demo: play.unity.com — Post-Combustion CO₂ Carbon Capture.

StockWatch iOS — Swift, SwiftUI, Swift Charts 2025

- Built a SwiftUI watchlist app supporting 55+ instruments across US, German Xetra, UK, Swiss, and Danish exchanges.
- Integrated live Yahoo Finance API quotes, interactive Swift Charts with eight time ranges, and in-app news feed.
- Implemented English/German runtime localisation, dark/light/system theming, and VoiceOver-accessible rows.
- MVVM architecture with protocol-based dependency injection and 11 XCTest unit test cases.
- Repository: github.com/hashwanthghanta/stockwatch-ios.

CO₂ Carbon Capture Data Dashboard — C#, Data Visualisation 2025

- Built a data-driven desktop dashboard visualising simulation axis data from the post-combustion CO₂ capture process.
- Integrated real-time data rendering with a clean analytical UI for capture-process stage review.
- Repository: github.com/hashwanthghanta/co2-carbon-extract.

Personal Portfolio Website — HTML5, CSS3, JavaScript 2025

- Designed and shipped a responsive single-page portfolio hosted on GitHub Pages with bilingual EN/DE toggle.
- Implemented custom CSS animations, dark/light theme, EmailJS contact form, and WCAG 2.2 AA accessibility patterns.
- Live site: hashwanthghanta.github.io.

IoT Smart Helmet and Bike Safety System — Final Year Project 2024

- Engineered a GPS and alcohol-sensor-equipped smart helmet that detects unsafe riding and alerts emergency contacts.
- Targeted motorcycle fatality reduction through automated real-time safety enforcement.

IoT-Based Ambulance Assistance System — VIT-AP University 2023

- Designed an intelligent IoT traffic-junction system granting priority clearance to emergency vehicles.
- Prototype implemented with microcontrollers, real-time sensor detection, and relay-based signal control.

Arduino Blind Spot Detector — VIT-AP University 2022

- Developed an ultrasonic-sensor blind-spot alert system for vehicles to reduce collision risk.

CERTIFICATIONS AND TRAINING

- Unity Certified Associate — Game Developer (in progress)
- Python for Data Science — IBM via Coursera
- Electronics and IoT Fundamentals — NPTEL
- Data Visualisation: Empowering Business with Effective Insights — Tata Group via Forage, 2024
- Career Essentials in Business Analysis — Microsoft and LinkedIn Learning, 2024
- Introduction to Business Analytics: Communicating with Data — Coursera

- IoT Wireless and Cloud Computing Emerging Technologies — Coursera
 - Introduction and Programming with IoT Boards — Coursera
 - Project Management Foundations — Coursera
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LEADERSHIP AND ACTIVITIES

Volunteer Head — VITOPIA 2024, VIT-AP University Annual Cultural Festival, India 2024

- Led the largest annual university event with over 1,000 participants including students and guest performers.
 - Coordinated logistics, volunteer teams, event scheduling, and live activity management across cultural and technical tracks.
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LANGUAGES

- English: Professional working proficiency (C1)
- German: A2 — actively enrolled in language course at OVGU Magdeburg
- Telugu / Hindi: Native